

# A Structural Model of Rational Addiction

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## 摘要

This paper develops a unified structural model of rational addiction that incorporates two complementary innovations into the classic Becker – Murphy (1988) framework. First, quadratic adjustment frictions capture the physiological and psychological costs of changing consumption habits. Second, a novel endogenous exit mechanism, modeled as an exponential utility component, reframes abstinence from a pure welfare loss into active asset management. We derive optimality conditions via dynamic programming, characterize multiple steady states, and conduct comparative static analysis. A key finding is that the interaction between adjustment frictions and the exit mechanism produces complex roots in the Euler system, thereby interpreting oscillatory (relapse-like) consumption paths as optimal behavior of fully forward-looking agents. The linearized Euler equation connects the theory with survey-based estimation and remains fully compatible with Chaloupka (1991) and Becker et al. (1994).

**Keywords:** rational addiction, habit formation, adjustment frictions, exit mechanism, relapse, dynamic programming.

**JEL Codes:** D11, D90, I12.

## 1 Introduction

The economic analysis of addiction has been decisively shaped by the rational addiction model of Becker and Murphy (1988). In contrast to views that treat addicts as impulsive or irrational, Becker and Murphy argue that even severe addictive behaviors can be understood as the outcome of consistent, forward-looking utility maximization. The key mechanism is reinforcement: past consumption raises the marginal utility of current consumption, making the two adjacent complements. This generates multiple steady states, cold-turkey quitting, and stronger demand responsiveness to permanent price changes than to temporary ones.

The existing literature has extended the Becker – Murphy framework along several dimensions. Orphanides and Zervos (1995) introduce uncertainty, showing that addiction can arise from misperception of personal susceptibility. Gruber and Koszegi (2001) relax the assumption of full rationality by introducing present bias to explain over-consumption. This paper is complementary to the above work: while maintaining full rationality, it provides a welfare-consistent explanation for oscillatory recovery paths through adjustment frictions and an endogenous exit mechanism.

We extend the baseline model along two dimensions. First, drawing on the habit formation literature (Fuhrer 2000, Carroll 2000), we introduce quadratic adjustment frictions that penalize changes in consumption. Second, we introduce an endogenous exit mechanism: an exponential utility component that makes the act of abstinence itself yield utility, reframing exit from an externally imposed constraint to an endogenous asset portfolio choice.

The interaction of these two features yields a new theoretical result: under specific parameter configurations, the characteristic roots of the linearized Euler equation become complex, implying oscillatory dynamics. Relapse is no longer a failure of willpower but an optimal response to dynamic frictions, thus providing a welfare-consistent rationalization of non-monotonic recovery paths.

The paper is structured as follows: Section 2 presents the structural model; Section 3 characterizes steady states and comparative statics; Section 4 contrasts rational and myopic addiction; Section 5 states the core propositions; Section 6 derives the linearized demand equation; Section 7 discusses policy implications; Section 8 outlines the empirical strategy; Section 9 concludes.

## 2 The Structural Model

### 2.1 Dynamics of Addiction Capital Stock

Following Becker and Murphy (1988), let  $A_t \geq 0$  denote the stock of addiction capital at the beginning of period  $t$ . The stock evolves according to:

$$A_{t+1} = (1 - \delta)A_t + c_t, \quad \delta \in (0, 1) \tag{1}$$

where  $c_t \geq 0$  is current consumption and  $\delta$  is the depreciation rate. Equation (1) incorporates tolerance (consumption augments the stock) and depreciation (the stock decays without consumption).

## 2.2 Instantaneous Utility

The per-period utility function comprises three components:

$$U_t = u(c_t, A_t) + g(c_t, A_t) - \phi(c_t, c_{t-1}) \quad (2)$$

**Direct utility.** We adopt a Cobb – Douglas form:

$$u(c_t, A_t) = c_t^\alpha A_t^\beta, \quad \alpha, \beta > 0 \text{ Direct utility satisfies the Cobb – Douglas constraint } \alpha + \beta = 1, \alpha \in (0, 1) \quad (3)$$

**Remark 1 (Economic meaning and limitations of  $\alpha + \beta = 1$ )** *The constraint  $\alpha + \beta = 1$  implies that direct utility  $u(c_t, A_t)$  is homogeneous of degree one in consumption and addiction capital; i.e., if  $(c_t, A_t)$  are scaled up proportionally, utility increases proportionally. This has a clear economic meaning in the addiction context: the agent’s satisfaction is determined solely by the relative ratio  $c_t/A_t$ , not by absolute levels, consistent with the concept of “tolerance” in the addiction literature — the agent needs more consumption at high addiction stocks to achieve the same satisfaction (Becker and Murphy 1988).*

*However, this constraint also rules out nonlinear scale effects of consumption on utility. If we relax it to  $\alpha + \beta \neq 1$ , then  $\alpha + \beta < 1$  corresponds to a risk-averse addict (decreasing returns to scale), and  $\alpha + \beta > 1$  to a deeply addicted individual with strong reinforcement (increasing returns). The choice  $\alpha + \beta = 1$  here balances analytical tractability and realism: the constraint reduces the steady-state conditions (10) – (12) to a single equation in  $c^*$ , substantially simplifying comparative statics, while numerical examples ( $\alpha = 0.4$ ,  $\beta = 0.6$ ) indicate local robustness of the conclusions near this constraint. The case  $\alpha + \beta \neq 1$  is left for future research.*

The cross-partial derivative  $\partial^2 u / \partial c_t \partial A_t = \alpha \beta c_t^{\alpha-1} A_t^{\beta-1} > 0$  confirms adjacent complementarity: a larger addiction stock raises the marginal utility of current consumption, the hallmark of reinforcement (Stigler and Becker 1977, Becker and Murphy 1988).

**Exit mechanism utility.** We model the utility generated by exiting (abstaining) through an exponential function of current consumption, anchored by the current addiction stock:

$$g(c_t, A_t) = \eta e^{-kc_t} - \eta e^{-kA_t}, \quad \eta, k > 0 \quad (4)$$

Here  $\eta$  controls the scale of the exit mechanism, and  $k$  reflects the physiological and psychological sensitivity to withdrawal. The elasticity of the exit mechanism,  $\varepsilon = -kc_t e^{-kc_t} / (e^{-kc_t} - e^{-kA_t})$ , is independent of  $\eta$  and tends to infinity as  $c_t$  approaches  $A_t$ , capturing the intense withdrawal shock at the onset of abstinence. A second-order Taylor expansion around  $c_t = 0$  yields:

$$g(c_t) \approx \eta(1 - kc_t + (kc_t)^2/2) - \eta e^{-kA_t} \quad (4')$$

The functional form (4) is chosen based on four considerations.

**First, neurobiological foundations.** The neuroscience of addiction indicates that during withdrawal, the activation intensity of brain reward circuits (particularly nucleus accumbens dopamine signals) increases convexly with the degree of dependence (Koob and Volkow 2010). The convexity of the exponential function  $e^{-kc_t}$  captures precisely this feature: the marginal exit utility from reducing one unit of consumption is far larger at high addiction stocks than at low stocks, consistent with clinical observations that acute withdrawal symptoms intensify nonlinearly with dependence severity.

**Second, compatibility with behavioral economics.** Exponential exit utility is compatible with the convexity of the loss domain in prospect theory (Kahneman and Tversky 1979)—withdrawal can be viewed as a “loss” of an existing consumption habit, with increasing marginal pain. The parameter  $k$  therefore has a clear behavioral interpretation: it measures an individual’s sensitivity to withdrawal loss, and can be mapped onto addiction severity scales (e.g., DSM-5 dependence scores).

**Third, the simplest nonlinear generalization of a linear benchmark.** As  $k \rightarrow 0$ , the Taylor expansion gives  $g(c_t, A_t) \approx \eta k(A_t - c_t)$ , degenerating to a linear exit benefit. Among all function families that satisfy convexity, the boundary condition  $g(A_t, A_t) = 0$ , and the linear degeneracy constraint, the exponential function is the member that introduces the fewest additional parameters (only  $k, \eta$ ), making it the simplest nonlinear generalization of the linear benchmark.

**Fourth, natural boundary conditions.** When  $c_t = A_t$ , we have  $g(A_t, A_t) = 0$ , meaning that the exit mechanism generates no utility if consumption is not reduced, consistent with the intuition that “only actual reductions in consumption yield abstinence benefits.” Moreover,  $g$  is monotonically decreasing in  $c_t$ , ensuring that exit benefits increase monotonically as consumption falls, avoiding a non-monotonic incentive structure.

The exponential exit utility herein is consistent with the tradition of modeling addiction nonlinearities in the literature, while offering a clearer economic interpretation through the boundary condition  $g(A_t, A_t) = 0$  (Orphanides and Zervos 1995).

**Adjustment frictions.** Inspired by Fuhrer (2000), we model the costs of changing consumption as a quadratic penalty:

$$\phi(c_t, c_{t-1}) = \frac{\gamma}{2}(c_t - c_{t-1})^2, \quad \gamma > 0 \quad (5)$$

As  $\gamma \rightarrow 0$ , the model reduces to the frictionless Becker – Murphy benchmark; as  $\gamma \rightarrow \infty$ , consumption becomes completely rigid.

Combining (3) – (5), the complete instantaneous utility is:

$$U_t = c_t^\alpha A_t^\beta + \eta e^{-kc_t} - \eta e^{-kA_t} - \frac{\gamma}{2}(c_t - c_{t-1})^2 \quad (6)$$

## 2.3 Dynamic Programming Problem

The agent maximizes discounted lifetime utility. The value function satisfies:

$$V(A_t, c_{t-1}) = \max_{c_t} \{U_t + \beta_d V(A_{t+1}, c_t)\} \quad (7)$$

subject to (1), where  $\beta_d \in (0, 1)$  is the subjective discount factor. Additionally, consumption satisfies the budget constraint  $c_t \in [0, \bar{c}]$ , where  $\bar{c}$  is determined by current disposable income; the state space is thus compact.

## 2.4 Optimality Conditions

**First-order condition (FOC).** Differentiating the right-hand side of (7) with respect to  $c_t$  and setting it to zero yields:

$$\alpha c_t^{\alpha-1} A_t^\beta - \eta k e^{-kc_t} - \gamma(c_t - c_{t-1}) + \beta_d \gamma \mathbb{E}_t[c_{t+1} - c_t] + \beta_d \mathbb{E}_t[\lambda_{t+1}] = 0 \quad (8)$$

where  $\lambda_{t+1} = \partial V(A_{t+1}, c_t) / \partial A_{t+1}$  is the shadow price of the addiction stock. The five terms represent: (i) the direct marginal utility of increasing consumption; (ii) the marginal disutility from the exit mechanism; (iii) the current-period adjustment cost; (iv) the discounted saving of the next period's adjustment cost; (v) the discounted shadow value of the stock.

**Envelope condition.** Applying the envelope theorem to (7) with respect to  $A_t$ :

$$\lambda_t = \beta_d \cdot \alpha c_t^\alpha A_t^{\beta-1} + \beta_d (1 - \delta) \lambda_{t+1} \quad (9)$$

## 2.5 Summary: Structural Equation System

Under rational expectations, the equilibrium path is fully described by (1), (8), and (9). As  $k \rightarrow 0$ ,  $\eta \rightarrow 0$ ,  $\gamma \rightarrow 0$ , the system reduces to the frictionless Becker – Murphy model.

# 3 Steady-State Analysis and Comparative Statics

## 3.1 Steady-State Conditions

Assume all variables are time-invariant and denote the steady state by  $(c^*, A^*, \lambda^*)$ . Setting  $c_t = c_{t-1} = c_{t+1} = c^*$  and  $A_t = A^*$  in (1), (8), and (9):

$$A^* = c^* / \delta \quad (10)$$

$$\alpha (c^*)^{\alpha-1} (A^*)^\beta - \eta k e^{-kc^*} + \beta_d \lambda^* = 0 \quad (11)$$

$$\lambda^* = \frac{\beta_d (c^*)^\alpha (A^*)^{\beta-1}}{1 - \beta_d (1 - \delta)} \quad (12)$$

Substituting (10) into (11) – (12) yields a single steady-state condition in  $c^*$ . The exit mechanism enters (11) via the additional concave term  $-\eta k e^{-kc^*}$ , which decreases with  $c^*$  at low consumption levels, thereby tilting the system toward a lower equilibrium consumption relative to the baseline Becker – Murphy model.

**Multiple steady states.** The system generally permits two stable steady states: a low-consumption equilibrium  $c_L^*$  and a heavy-addiction equilibrium  $c_H^*$ , separated by an unstable threshold  $c^\dagger$ . Relative to the frictionless model, the exit mechanism compresses the basin of attraction of  $c_H^*$ , implying that smaller perturbations can shift the agent from heavy addiction back to low consumption.

### 3.2 Comparative Statics via the Implicit Function Theorem

Define  $F(c^*; \delta, \gamma, \beta_d, k, \eta) = 0$  from the steady-state equations. When  $dF/dc^* \neq 0$ , the implicit function theorem yields:

$$\frac{dc^*}{d\theta} = -\frac{\partial F/\partial \theta}{\partial F/\partial c^*}, \quad \theta \in \{\delta, \gamma, \beta_d, \alpha, k, \eta\} \quad (13)$$

表 1: Comparative static effects on steady-state consumption  $c^*$

| Parameter          | Economic meaning            | Sign and intuition of $dc^*/d\theta$                      |
|--------------------|-----------------------------|-----------------------------------------------------------|
| $\delta \uparrow$  | Faster stock depreciation   | –; stock resets quickly, cumulative reinforcement w       |
| $\gamma \uparrow$  | Higher adjustment frictions | +; quitting is more costly, addiction is stickie          |
| $\beta_d \uparrow$ | Greater patience            | –; future health costs fully internalized                 |
| $\alpha \uparrow$  | Stronger direct utility     | +; consumption yields higher benefits                     |
| $k \uparrow$       | Higher exit sensitivity     | –; abstinence more attractive, depresses high-consumption |
| $\eta \uparrow$    | Larger exit mechanism scale | –; higher utility from exit reduces $c^*$                 |

## 4 Rational Addiction and Myopic Addiction

A myopic agent sets  $\beta_d = 0$  in the FOC (8), obtaining:

$$\alpha c_t^{\alpha-1} A_t^\beta - \eta k e^{-kc_t} - \gamma(c_t - c_{t-1}) = 0 \quad (14)$$

Comparing (8) and (14), the rational model adds two terms: (i)  $\beta_d \gamma \mathbb{E}[c_{t+1} - c_t]$ : the rational agent anticipates future adjustment costs and smooths consumption; (ii)  $\beta_d \mathbb{E}[\lambda_{t+1}]$ : the rational agent internalizes the impact of current consumption on future cravings.

These terms generate the model's key empirical prediction: current consumption depends not only on the past ( $c_{t-1}$ ) but also on expected future consumption ( $\mathbb{E}[c_{t+1}]$ ).

A significant positive coefficient on the future term is an empirical hallmark of forward-looking behavior (Chaloupka 1991, Becker et al. 1994).

## 5 Core Propositions

**Proposition 1 (Hedging Logic).** The exit mechanism provides an endogenous, continuously available utility flow that transforms abstinence from a “pure welfare loss” into active asset management. The parameter  $k$  ensures that the marginal value of exit accurately reflects the physiological and psychological reality of withdrawal: the gain from reducing consumption is steepest precisely when consumption is closest to the current addiction stock  $A$ . This provides the agent with a dynamic hedging instrument against the accumulation of addiction capital.

**Proposition 2 (Rational Relapse).** Due to the interaction of the adjustment friction coefficient  $\gamma$  and the curvature of the exit mechanism  $k$ , the characteristic equation of the linearized Euler system yields complex roots under specific parameter configurations (when both  $\gamma$  and  $k$  exceed thresholds determined by  $\alpha, \beta, \delta, \beta_d$ ). This leads to oscillatory, relapse-like consumption paths, where the agent strategically bounces back to smooth the sum of friction and exit utility losses. Thus, relapse is an optimal rational path, not a failure of willpower.

To characterize local dynamics, we linearize the optimality conditions around the steady state. Incorporating the law of motion for addiction capital, we can delineate three regimes of optimal consumption paths.

[Three Regimes of Optimal Consumption Paths] Define the second-order partial derivative of the utility function with respect to consumption at the steady state  $(\bar{c}, \bar{A})$  as

$$H_c = -\alpha\beta\bar{c}^{\alpha-2}\bar{A}^\beta + \eta k^2 e^{-k\bar{c}}, \quad (*)$$

where the first term (standard concavity) is always negative, and the second term (exit shock) is always positive; their relative strength determines the regime. Further define

$$\Omega = \gamma(1 + \beta_d) - H_c, \quad \Delta = \Omega^2 - 4\beta_d\gamma^2.$$

Then the optimal consumption path  $\tilde{c}_t$  falls strictly into three regimes:

**Regime I (Rational Abstinence: Monotone Convergence).** If the exit shock is weaker than standard concavity:

$$\eta k^2 e^{-k\bar{c}} < \alpha\beta\bar{c}^{\alpha-2}\bar{A}^\beta,$$

i.e.,  $H_c < 0$ , then  $\Delta > 0$  and the characteristic equation possesses a saddle-point structure with  $\lambda_1 \in (0, 1)$  and  $\lambda_2 > 1$ . The rational agent monotonically converges to the steady state along the stable saddle path, i.e., gradual abstinence.

**Regime II (Rational Relapse: Bounded Oscillations).** If the exit shock pushes  $H_c$  into the following window:

$$\gamma(1 - \sqrt{\beta_d})^2 < -\alpha\beta \bar{c}^{\alpha-2} \bar{A}^\beta + \eta k^2 e^{-k\bar{c}} < \gamma(1 + \sqrt{\beta_d})^2,$$

i.e.,  $\Delta < 0$ , then the characteristic equation has complex conjugate roots  $\lambda_{1,2} = \rho e^{\pm i\theta}$ , with modulus  $\rho = \beta_d^{-1/2} > 1$ . After imposing the budget constraint  $c_t \in [0, \bar{c}]$ , the state space becomes compact, and the bounded oscillation property of the path is stated as a conjecture, supported by the numerical example below.

**Regime III (Uncontrolled Addiction: Complete Divergence).** If the exit shock is too strong, making  $H_c$  exceed the upper bound of Regime II:

$$-\alpha\beta \bar{c}^{\alpha-2} \bar{A}^\beta + \eta k^2 e^{-k\bar{c}} \geq \gamma(1 + \sqrt{\beta_d})^2,$$

or if  $H_c$  falls into the transition band  $[0, \gamma(1 - \sqrt{\beta_d})^2]$ , then  $\Delta \geq 0$  and both characteristic roots have modulus greater than 1; the path diverges without bound, i.e., addiction spirals completely out of control.

The characteristic equation is  $\beta_d \gamma \lambda^2 - \Omega \lambda + \gamma = 0$ . By Vieta's formulas:

$$\lambda_1 \lambda_2 = \frac{1}{\beta_d} > 1, \quad \lambda_1 + \lambda_2 = \frac{\Omega}{\beta_d \gamma}. \quad (\text{V})$$

Evaluating the characteristic polynomial at  $\lambda = 1$ :

$$P(1) = \gamma(1 + \beta_d) - \Omega = H_c = -\alpha\beta \bar{c}^{\alpha-2} \bar{A}^\beta + \eta k^2 e^{-k\bar{c}}. \quad (\text{P1})$$

*Regime I.* When  $H_c < 0$ , we have  $P(1) < 0$ ,  $P(0) = \gamma > 0$ ,  $P(+\infty) = +\infty$ . The intermediate value theorem guarantees one root in  $(0, 1)$  and one in  $(1, +\infty)$ . Since the product of roots is positive, they have the same sign, hence  $\lambda_1 \in (0, 1)$  and  $\lambda_2 > 1$ .

Moreover, by the inequality  $1 + \beta_d \geq 2\sqrt{\beta_d}$ ,  $H_c < 0$  implies  $\Omega > \gamma(1 + \beta_d) \geq 2\sqrt{\beta_d} \gamma$ , so  $\Delta > 0$  automatically holds.  $\square_1$

*Regime II.*  $\Delta < 0$  is equivalent to  $|\Omega| < 2\sqrt{\beta_d} \gamma$ . Expanding and rearranging yields

$$\gamma(1 - \sqrt{\beta_d})^2 < H_c < \gamma(1 + \sqrt{\beta_d})^2.$$

The characteristic roots are complex conjugates, with modulus given by Vieta:

$$\rho = \sqrt{\lambda_1 \lambda_2} = \frac{1}{\sqrt{\beta_d}} > 1.$$

The linearized system itself diverges, but the constraint  $c_t \geq 0$  reflects the path back into the positive quadrant, generating bounded oscillations. The angular frequency of oscillation is

$$\omega = \arctan\left(\frac{\sqrt{|\Delta|}}{\Omega}\right). \square_2$$



*Regime III (subcase 3a).* If  $H_c \geq \gamma(1 + \sqrt{\beta_d})^2$ , then  $\Omega \leq -2\sqrt{\beta_d}\gamma < 0$ , and both roots are negative real numbers. Evaluating at  $\lambda = -1$ :

$$P(-1) = \gamma(1 + \beta_d) + \Omega \leq \gamma(1 - \sqrt{\beta_d})^2 \geq 0,$$

hence both roots satisfy  $\lambda \leq -1$ , i.e.,  $|\lambda_{1,2}| \geq \beta_d^{-1/2} > 1$ , and the path diverges.

*Regime III (subcase 3b).* If  $0 \leq H_c \leq \gamma(1 - \sqrt{\beta_d})^2$ , then  $\Omega \geq 2\sqrt{\beta_d}\gamma > 0$ , and both roots are positive real numbers. Since  $P(1) = H_c \geq 0$  and  $P(0) > 0$ , neither root lies in  $(0, 1)$ ; combined with the product  $= 1/\beta_d > 1$ , both roots exceed 1, and the path diverges monotonically.  $\square_3$

**Remark 2** After imposing  $\alpha + \beta = 1$ ,  $H_c$  exhibits a clear offsetting structure: the standard concavity term  $-\alpha\beta\bar{c}^{\alpha-2}\bar{A}^\beta$  (determined solely by preference parameters) and the exit shock term  $\eta k^2 e^{-k\bar{c}}$  (determined by withdrawal sensitivity) oppose each other, and their relative strength uniquely determines the regime. In particular, the exit shock term  $\eta k^2 e^{-k\bar{c}}$  attains its maximum at  $k = 2/\bar{c}$ , and thereafter decays monotonically to zero as  $k$  increases. This implies the existence of an optimal intervention intensity: too low a  $k$  fails to produce relapse oscillations, while too high a  $k$  dissipates the shock; neither can realize the rational relapse regime. This provides a theoretical basis for the dosing design of medication-assisted treatment (MAT).

**Discussion.** Proposition 2 offers a welfare-consistent rationalization of oscillatory recovery trajectories within a fully rational framework. The frequency of oscillation depends on the imaginary part of the complex roots: it increases with  $\gamma$  (stronger friction lengthens the cycle) and decreases with  $k$  (more sensitive exit utility subjects the agent to stronger corrective incentives sooner, shortening the cycle). This implies that medication-assisted treatment programs that lower  $\gamma$  can compress the relapse cycle independently of punitive taxes.

## Numerical Example

To illustrate the economic meaning of the three regimes, we fix the baseline parameters  $\alpha = 0.4$ ,  $\beta = 0.6$ ,  $\delta = 0.2$ ,  $\beta_d = 0.95$ ,  $\gamma = 1.0$ , and impose  $\alpha + \beta = 1$ . Under these parameters, the steady-state constant is

$$M \equiv \alpha\delta^{-\beta} + \frac{\beta_d^2 \delta^{1-\beta}}{1 - \beta_d(1 - \delta)} = 0.4 \times 0.2^{-0.6} + \frac{0.9025 \times 0.2^{0.4}}{0.24} \approx 3.026, \quad (1)$$

and steady-state consumption is uniquely determined by  $\eta k e^{-kc^*} = M$ , i.e.,  $c^* = k^{-1} \ln(\eta k / M)$ . We fix  $c^* = 0.50$  ( $A^* = c^*/\delta = 2.50$ ) and adjust  $\eta$  so that the steady-state equation holds exactly in each regime (see Table 2).

Using  $\alpha + \beta = 1$  to simplify, the second-order derivative of utility with respect to consumption at the steady state reduces to

$$H_c = -\frac{\alpha\beta\delta^{-\beta}}{c^*} + kM = -\frac{0.457}{0.5} + 3.026k = -0.914 + 3.026k. \quad (2)$$

The regime boundaries are  $\gamma(1 - \sqrt{\beta_d})^2 \approx 0.001$  and  $\gamma(1 + \sqrt{\beta_d})^2 \approx 3.899$ .

表 2: Numerical examples for the three regimes ( $c^* = 0.50$ ,  $A^* = 2.50$ )

| Regime                       | $k$  | $\eta$ | $H_c$  | $\Delta$ | Path behavior        |
|------------------------------|------|--------|--------|----------|----------------------|
| I (Rational abstinence)      | 0.20 | 16.72  | -0.309 | $> 0$    | Monotone convergence |
| II (Rational relapse)        | 0.80 | 5.64   | 1.507  | -3.604   | Bounded oscillation  |
| III (Uncontrolled addiction) | 1.60 | 4.21   | 3.928  | $> 0$    | Unbounded divergence |

**Detailed calculation for Regime II.** Taking  $k = 0.80$ ,  $\eta = 5.64$ , verify  $\eta ke^{-kc^*} = 5.64 \times 0.8 \times e^{-0.4} \approx 3.026 = M$ . Then

$$\Omega = \gamma(1 + \beta_d) - H_c = 1.95 - 1.507 = 0.443, \quad (3)$$

$$\Delta = \Omega^2 - 4\beta_d\gamma^2 = 0.196 - 3.800 = -3.604 < 0. \quad (4)$$

The characteristic roots are complex conjugates:

$$\lambda_{1,2} = \frac{\Omega \pm i\sqrt{|\Delta|}}{2\beta_d\gamma} = \frac{0.443 \pm 1.899i}{1.9} \approx 0.233 \pm 0.999i, \quad (5)$$

with modulus  $\rho = \sqrt{0.233^2 + 0.999^2} \approx 1.026 = \beta_d^{-1/2}$  (consistent with Vieta's formulas), and argument  $\theta = \arctan(0.999/0.233) \approx 1.344$  radians, corresponding to an oscillation period of  $T = 2\pi/\theta \approx 4.7$  periods. Figure 1 displays the simulated path starting from an initial perturbation  $c_0 = 0.80$ : the amplitude is slowly amplified by  $\rho > 1$  until the constraint  $c_t \geq 0$  reflects the path back into the positive half-plane, generating bounded oscillations — a numerical manifestation of rational relapse.

**Conjecture (Global Bounded Oscillations).** Under the parameter configuration of Regime II, if the budget constraint  $c_t \in [0, \bar{c}]$  is further imposed, the state space becomes compact, and the global behavior of the optimal path on this compact set is consistent with the bounded oscillations shown in Figure 1. A rigorous proof would require constructing a positively invariant set for the original nonlinear system; this is stated as a conjecture, with numerical simulations providing supporting evidence.

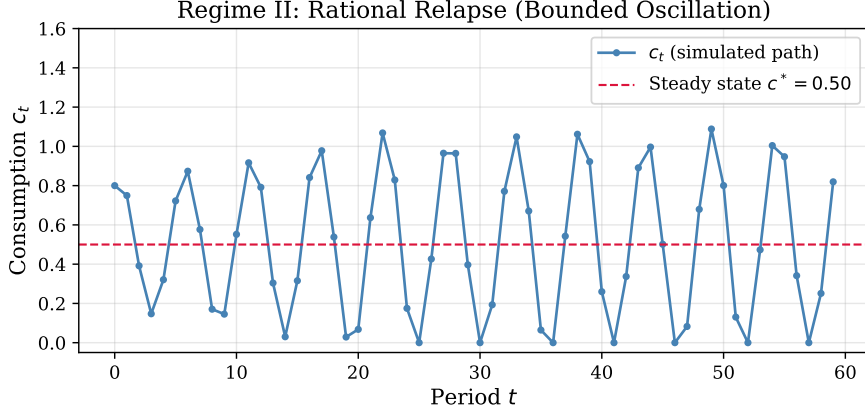
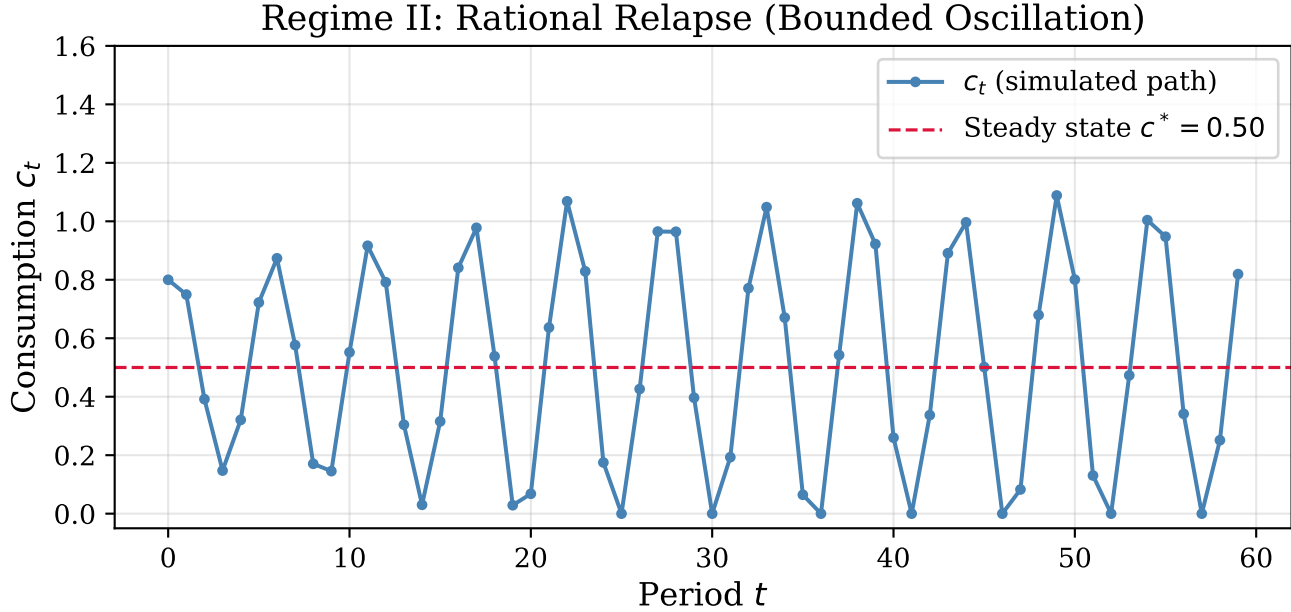


图 1: Simulated consumption path for Regime II ( $k = 0.80$ ,  $\eta = 5.64$ ). The red dashed line is the steady state  $c^* = 0.50$ ; the blue solid line is the optimal path starting from  $c_0 = 0.80$  subject to the non-negativity constraint, exhibiting bounded oscillations (rational relapse). Parameters:  $\alpha = 0.4$ ,  $\beta = 0.6$ ,  $\delta = 0.2$ ,  $\beta_d = 0.95$ ,  $\gamma = 1.0$ .



## 6 Linearization and an Estimable Euler Equation

### 6.1 First-Order Taylor Expansion

The structural system is nonlinear and cannot be estimated directly. Following Chaloupka (1991) and Becker et al. (1994), we perform a first-order Taylor expansion of the FOC (8) around the steady state  $(c^*, A^*)$ . Denote  $\tilde{c}_t = c_t - c^*$  and  $\tilde{A}_t = A_t - A^*$ . The

first-order expansion of the nonlinear term  $h(c_t, A_t) = \alpha c_t^{\alpha-1} A_t^\beta - \eta k e^{-kc_t}$  is:

$$h(c_t, A_t) \approx h(c^*, A^*) + \left. \frac{\partial h}{\partial c} \right|^* \tilde{c}_t + \left. \frac{\partial h}{\partial A} \right|^* \tilde{A}_t \quad (15)$$

where the steady-state partial derivatives are:

$$\left. \frac{\partial h}{\partial c} \right|^* = \alpha(\alpha - 1)(c^*)^{\alpha-2} (A^*)^\beta - \eta k^2 e^{-kc^*} \quad (16)$$

$$\left. \frac{\partial h}{\partial A} \right|^* = \alpha \beta (c^*)^{\alpha-1} (A^*)^{\beta-1} \quad (17)$$

## 6.2 Linearized Demand Equation

Substituting (15) into the linearized version of (8) and using the linearized stock equation  $\tilde{A}_t \approx (1 - \delta)\tilde{A}_{t-1} + \tilde{c}_{t-1}$ , we obtain the reduced-form demand equation:

$$c_t = \phi_0 + \phi_1 c_{t-1} + \phi_2 \mathbb{E}_t[c_{t+1}] + \phi_3 P_t + \varepsilon_t \quad (18)$$

where  $P_t$  is the real price of the addictive good,  $\varepsilon_t$  is a zero-mean error term, and the reduced-form coefficients  $(\phi_0, \phi_1, \phi_2, \phi_3)$  are functions of the structural parameters  $(\alpha, \beta, \delta, \gamma, \beta_d, k, \eta)$ .

**Proposition 3 (Testable Predictions).** Under the rational addiction model:

1.  $\phi_1 > 0$  (reinforcement: past consumption raises current consumption);
2.  $\phi_2 > 0$  (forward-looking behavior: expected future consumption enters current demand);
3.  $\phi_3 < 0$  (downward-sloping demand);
4.  $\phi_2/\phi_1 = \beta_d$  (the ratio of the future coefficient to the past coefficient identifies the discount factor).

Condition (2) distinguishes the rational model from the myopic model ( $\phi_2 = 0$ ). The exit mechanism modifies the intercept  $\phi_0$  and the magnitudes of  $\phi_1$  and  $\phi_2$  through the steady-state value (16), but does not overturn any of these testable predictions, and thus remains fully empirically compatible with Chaloupka (1991) and Becker et al. (1994).

## 7 Policy Implications

**Permanent versus temporary price changes.** Because rational consumers are forward-looking, they respond to permanent price levels. The steady-state effect of a temporary tax increase of the same magnitude is smaller than that of a permanent tax

increase, providing a theoretical basis for durable fiscal instruments (e.g., a permanent increase in excise taxes).

**The policy role of  $\gamma$ .** Larger adjustment frictions  $\gamma$  raise the heavy-addiction steady state and lengthen the relapse cycle (Proposition 2). When  $\gamma$  is large, policies that prevent initiation (e.g., youth access restrictions, minimum age laws) are particularly effective at keeping agents within the basin of attraction of the low-consumption equilibrium. Medication-assisted treatment programs, by lowering  $\gamma$ , can simultaneously shift down the heavy-addiction steady state and compress the relapse cycle.

**Exit-targeting policies.** Proposition 1 implies that policy should focus on enhancing the exit mechanism rather than relying solely on punitive taxation. Raising  $\eta$  (the scale) through subsidized withdrawal assistance and counseling increases the steady-state abstinence incentive. Raising  $k$  (sensitivity) through graduated reduction programs makes the exit option more responsive to small reductions in consumption. This reorients the policy objective from punishing consumption to actively improving the utility of exit.

**Myopia and the case for intervention.** Gruber and Koszegi (2001) argue that present bias can cause agents to over-consume relative to their own long-run preferences. Our model yields a lower bound for optimal corrective taxation: the Pigouvian tax should at least internalize external costs, and an additional tax may be warranted if agents exhibit present bias.

## 8 Empirical Strategy

### 8.1 Estimation Equation

The empirical analysis applies equation (18) to survey data. Because  $\mathbb{E}_t[c_{t+1}]$  is unobservable, we follow the instrumental variable approach of Chaloupka (1991): future consumption  $c_{t+1}$  is used as a proxy for  $\mathbb{E}_t[c_{t+1}]$ , and lagged prices  $P_{t-1}$  serve as instruments for the endogenous variables  $c_{t-1}$  and  $c_{t+1}$ .

### 8.2 Questionnaire Design

The primary data source is an original questionnaire. Key variables include:

- Current consumption ( $c_t$ ): self-reported frequency and quantity;
- Past consumption ( $c_{t-1}$ ): consumption in the previous period;
- Price ( $P_t$ ): average expenditure per unit or local market price;
- Addiction stock proxy ( $A_t$ ): years of use, dependence severity scale scores;

- Exit behavior: self-reported quit attempts, use of cessation aids, relapse events — used as proxies for the exit parameters  $k$  and  $\eta$ ;
- Demographics: age, education, income — used to control for heterogeneity in  $\beta_d$ ,  $\alpha$ , and  $k$ .

### 8.3 Identifying Forward-Looking Behavior

The central identification challenge is to distinguish the rational model ( $\phi_2 > 0$ ) from the myopic model ( $\phi_2 = 0$ ). Following Becker et al. (1994), we estimate equation (18) with and without the future consumption term and conduct a Wald test of  $H_0 : \phi_2 = 0$ . A secondary test exploits Proposition 2: if relapse cycles are observed in panel data, and their frequency is positively correlated with  $\gamma$  (proxied by addiction severity) and negatively correlated with  $k$  (proxied by cessation aid use), this would provide additional support for the rational relapse mechanism.

## 9 Conclusion

This paper has developed a unified structural model that introduces quadratic adjustment frictions and an endogenous exit mechanism into the analysis of rational addiction. The main contributions include: (i) a clear derivation of optimality conditions via dynamic programming and the envelope theorem, integrating three utility components into a single coherent framework; (ii) a steady-state analysis showing that multiple equilibria arise naturally, with the exit mechanism compressing the basin of attraction of the heavy-addiction equilibrium; (iii) comparative static results linking the two new parameters  $(k, \eta)$  to intuitive economic predictions for steady-state consumption; (iv) the central theoretical result (Proposition 2) demonstrating that oscillatory relapse paths are the optimal response of fully rational agents to the interaction of adjustment frictions and exit-mechanism sensitivity; (v) a linearized Euler equation that bridges theory and empirics while remaining fully compatible with Chaloupka (1991) and Becker et al. (1994). Empirical analysis based on original survey data will test the predictions of forward-looking behavior ( $\phi_2 > 0$ ) and rational relapse. Parameter estimates will yield the implied discount factor  $\beta_d = \phi_2/\phi_1$  and the exit-mechanism parameters  $(k, \eta)$ , thereby providing a richer characterization of addictive behavior than existing studies.

The paper has several limitations. First, the model adopts a discrete-time framework; a continuous-time extension might yield more tractable analytical results. Second, the representative-agent assumption ignores heterogeneity among addicts, which could be crucial for policy evaluation. Third, the empirical analysis remains to be completed, and

the rational relapse prediction of Proposition 2 as well as the identification of structural parameters await data validation. These extensions are left for future research.

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